

CM0081 Formal Languages and Automata

Regular Expression in Haskell: An Introduction

Andrés Sicard-Ramírez

Universidad EAFIT

Semester 2024-1

Preliminaries

Conventions

- The number and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook (Hopcroft, Motwani and Ullman 2007).
- The natural numbers include the zero, that is, $\mathbb{N} = \{0, 1, 2, \dots\}$.
- The power set of a set A , that is, the set of its subsets, is denoted by $\mathcal{P}A$.

Introduction

- There are various libraries for handling regular expressions in `HASKELL`.
- POSIX (Portable Operating System Interface) is a family of standards specified for maintaining compatibility between operating systems.

Notation for Regular Expressions

POSIX	Textbook
ab	ab
$a b$	$a + b$
a^*	a^*
(a)	(a)
a^+	aa^*
$a?$	$a + \varepsilon$
$[abc]$	$a + b + c$
$.$	Any symbol

Demo

Examples

We shall use GHC 9.6.2, the libraries REGEX-POSIX 0.96.0.1 and REGEX-TDFA 1.3.2.1[†] and we shall see some examples from (O'Sullivan, Goerzen and Stewart 2008, Ch. 8)

[†]Hackage: <https://hackage.haskell.org/package/regex-posix> and <https://hackage.haskell.org/package/regex-tdfa>, respectively.

From the description of REGEX-BASE 0.94.0.2:[†]

This package does not provide the ability to do regular expression matching. Instead, it provides the type classes that constitute the abstract API that is implemented by regex- backends such as:*

- REGEX-POSIX
- REGEX-PARSEC
- REGEX-DFA
- REGEX-TDFA
- REGEX-PCRE

[†]<https://hackage.haskell.org/package/regex-base>.

From the description of REGEX-POSIX 0.96.0.1:[†]

Benchmarking shows the default regex library on many platforms is very inefficient. You might increase performance by an order of magnitude by obtaining LIBPCRE and REGEX-PCRE or LIBTRE and REGEX-TRE. If you do not need the captured substrings then you can also get great performance from REGEX-DFA. If you do need the capture substrings then you may be able to use REGEX-PARSEC to improve performance.

[†]<https://hackage.haskell.org/package/regex-posix-0.96.0.1/docs/Text-Regex-Posix.html>.

References

-  John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman [1979] (2007). Introduction to Automata Theory, Languages, and Computation. 3rd ed. Pearson Education (cit. on p. 2).
-  Bryan O'Sullivan, John Goerzen and Don Stewart (2008). Real World Haskell. O'Really Media, Inc. (cit. on p. 5).